

Praveen Kumar

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PROFILE

AI/ML Engineer specializing in LLM applications, Agentic RAG, AI guardrails, and production ML systems. Experienced in building and deploying LangGraph-based agents, RAG pipelines, multimodal VLM systems, and FastAPI services on AWS.

EXPERIENCE

Software Developer

July 2026 – Aug 2026

BharatOne, Pune

Remote

- Built and shipped a production chat application (Firestore/Auth/Cloud Functions) with end-to-end encrypted messaging and a caching layer, serving real users with low-latency delivery and native social-media sharing via a JWT-secured backend
- Designed and integrated an AI chatbot into the app with custom guardrails to sanitize inputs/outputs and block prompt-injection and abuse attempts, hardening the assistant against adversarial attacks in a live production environment

AI Developer-Intern

June 2026 – July 2026

Aigetai, Bengaluru

Hybrid

- Architected and deployed production-grade Retrieval-Augmented Generation (RAG) pipelines, integrating advanced AI gateways and custom routing logic for optimized LLM inference
- Engineered robust guardrails and comprehensive evaluation (evals) frameworks to ensure output safety, reliability, and alignment with enterprise quality standards

Artificial Intelligence Trainer (Freelance)

Dec 2024 – Feb 2025

Outlier, United States

Remote

- Improved LLM output quality across STEM domains through advanced prompt engineering, chain-of-thought reasoning, and RLHF evaluation workflows, ensuring technical accuracy and ethical alignment

PROJECTS

Multimodal-Guardrail | *FastAPI, Gemini Vision, ChromaDB, OpenCV, LangSmith, Docker, HuggingFace* Sep 2026

- Built a real-time multimodal guardrail microservice that intercepts spatial and temporal hallucinations in VLM video captioning before they reach downstream systems, running a parallel Dual-Judge Engine at **sub-450ms** end-to-end latency
- Prevented cascading hallucination (“memory poisoning”) across video streams by engineering a ChromaDB sliding-window buffer with BAAI/bge-small-en-v1.5 embeddings that auto-quarantines corrupted frames before they contaminate future context

Agentic Data Analyst | *E2B Code Interpreter v2, LangGraph, Docker, FastAPI, Nginx, Numpy, AWS EC2* Feb 2026

- Engineered a self-healing agentic pipeline using LangGraph’s stateful orchestration with conditional routing and error-reflection retries, enabling autonomous analysis from file ingestion to interactive Plotly dashboards, zero human-in-the-loop
- Designed a secure, multi-tenant execution engine with cloud-sandboxed micro-VMs and per-job isolation, then production-deployed the full stack on AWS EC2 as a single Dockerized service behind Nginx with auto-recovery and health monitoring

ACHIEVEMENTS

- National Finalist - 5G Lab Innovation Hackathon (2025):** Built AR smart glasses using 5G URLLC for real-time communication, selected among 1000 teams and awarded INR 1,00,000 funding for prototyping.

TECHNICAL SKILLS

Languages: C++, Python, JavaScript, HTML/CSS, SQL

AI & ML: RAG, GraphRAG, GenAI, AgenticAI, LLMs, NLP, TensorFlow, PyTorch, Scikit-learn, Pandas, NumPy

Observability & Gateways: LLM Evals, NeMo Guardrails, AWS Bedrock Guardrails, LangSmith, Portkey

Frameworks & Tools: LangChain, LangGraph, LlamaIndex, HuggingFace, FastAPI, Neo4j

Cloud & DevOps: AWS, Docker, Kubernetes, CI/CD (GitHub Actions), Git, Linux

EDUCATION

Central University of Haryana

Bachelor of Technology (B.Tech.) in Computer Science and Engineering

CGPA: 7.68/10.0

Mahendragarh, HR

Aug 2023 – July 2027